AI-Powered Healthcare Claims Automation System

Portfolio Case Study | Program Management + Business Analysis + AI + Application Operations

Designed as a fictional enterprise transformation program, this case study demonstrates how an AI-enabled healthcare claims platform can reduce manual processing, improve adjudication quality, strengthen fraud controls, and provide executive-level operational visibility. The project is intentionally structured to showcase strong program management, business analysis, change leadership, and healthcare technology fluency.

Business Problem

Claims organizations often struggle with long turnaround times, avoidable denials, manual rework, fragmented intake channels, and limited visibility across the claim lifecycle. These problems increase operating cost and make SLA management difficult.

Solution Overview

The proposed solution combines a claims automation engine, AI-driven requirements builder, FHIR/API integration layer, fraud analytics, and reporting dashboards. The target state supports intake, validation, classification, adjudication, exception handling, settlement, and continuous improvement.

My Role

Program Manager / BA Lead responsible for scope planning, stakeholder workshops, delivery governance, RAID management, UAT readiness, release reporting, change management, and adoption support across technology and operations.

Key Outcomes

The fictional program reduced claims processing time by 40%, improved adjudication accuracy by 30%, reduced denials and rework by 25%, and increased operational efficiency by 35% while maintaining strong governance and auditability.

AI Health Analytics Dashboard

Diagram included in the GitHub portfolio to visually explain the operating model and target architecture.

End-to-End Claims Workflow

Diagram included in the GitHub portfolio to visually explain the operating model and target architecture.

FHIR Integration Architecture

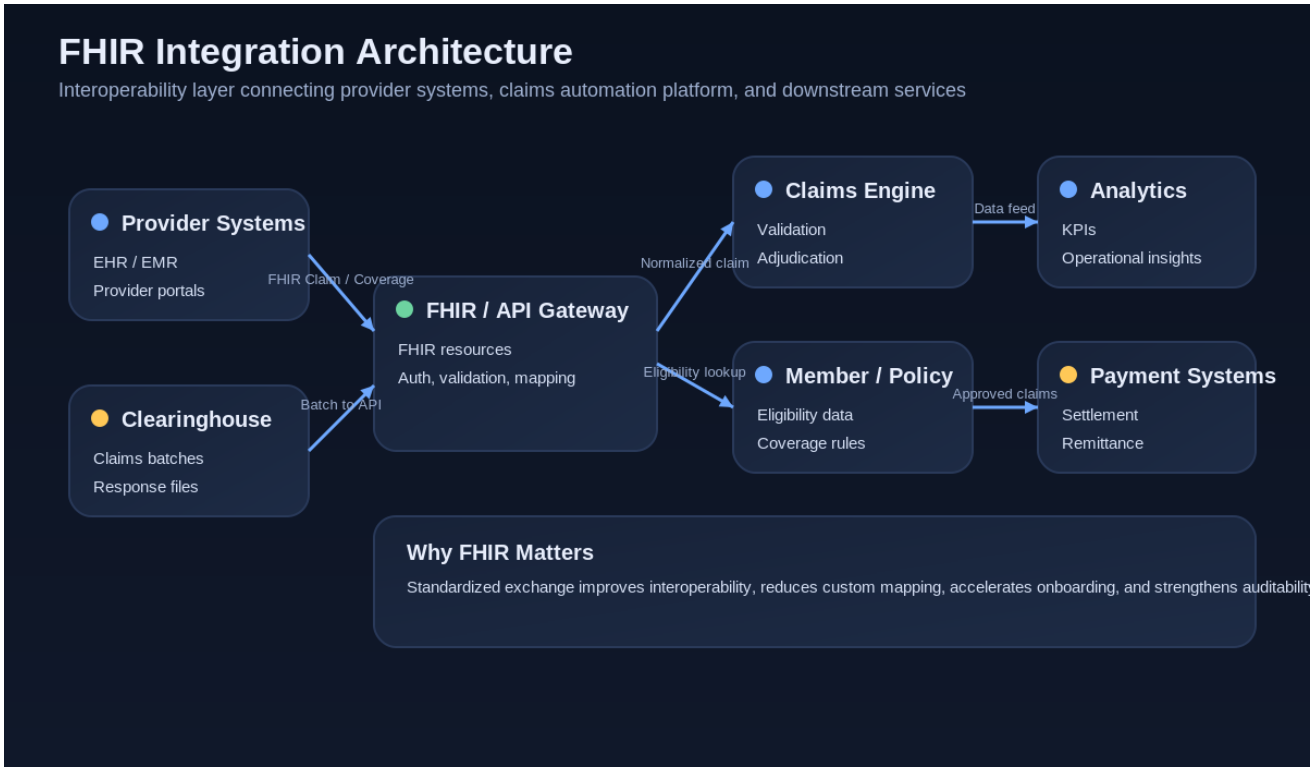


Diagram included in the GitHub portfolio to visually explain the operating model and target architecture.

AI Requirements Builder

Stakeholder inputs from workshops, SOPs, operational pain points, and policy requirements are transformed into draft BRD sections, user stories, acceptance criteria, dependency prompts, risk statements, and UAT scenarios. This positions AI as a documentation accelerator with human validation built into the process.

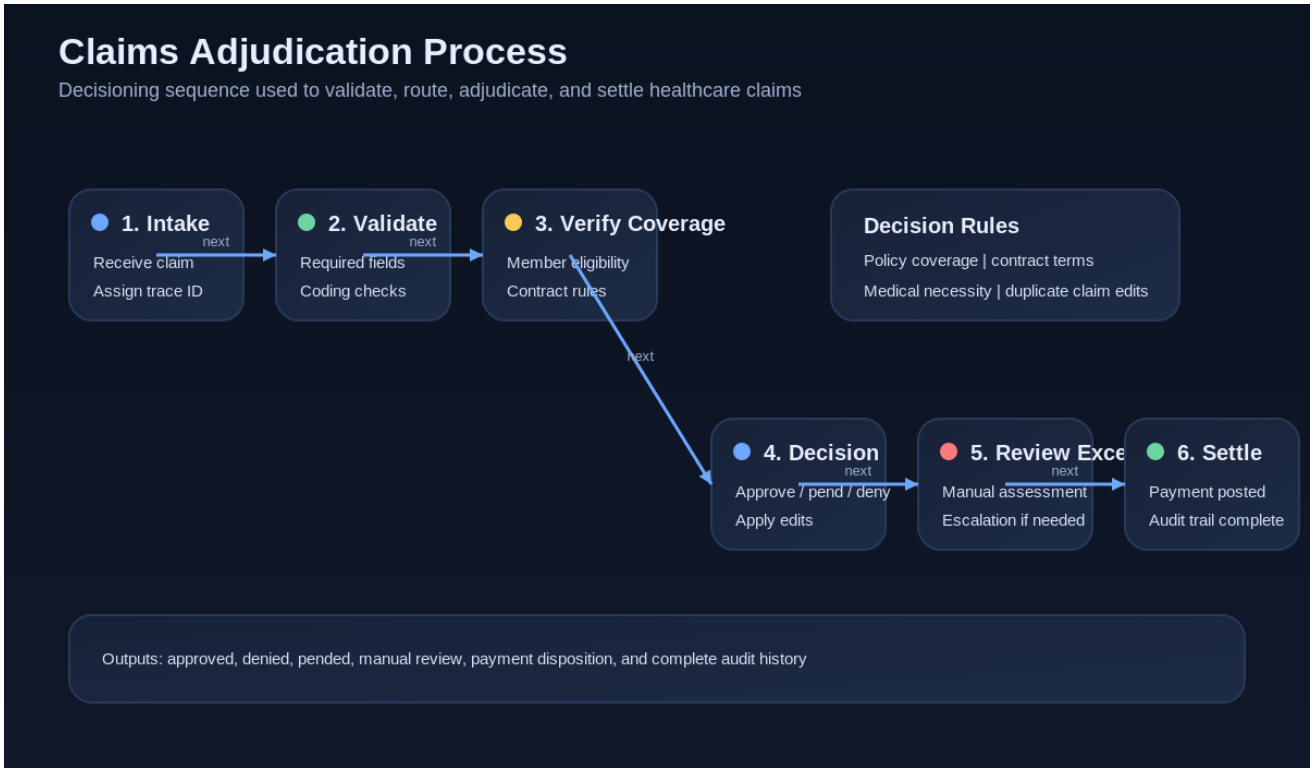
Claims Adjudication Model

The adjudication sequence begins with intake and traceability, followed by validation, eligibility checks, contract and policy logic, decisioning, exception routing, manual review where necessary, and settlement. Each stage includes controls that support compliance and operational audit needs.

Portfolio Value

This project is designed to help recruiters and hiring managers quickly understand capability in program delivery, business analysis, healthcare technology, digital transformation, and AI-enabled process modernization.

Claims Adjudication Process



AI Requirements Builder Flow

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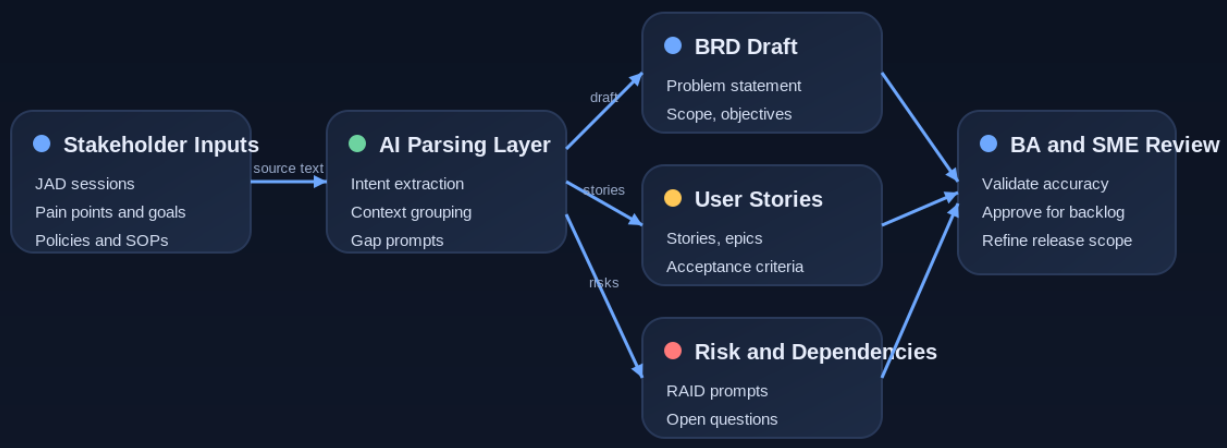
How stakeholder inputs are transformed into delivery-ready requirements and controls



Benefits: faster documentation cycles | higher consistency | stronger traceability | better handoff to delivery and testing teams